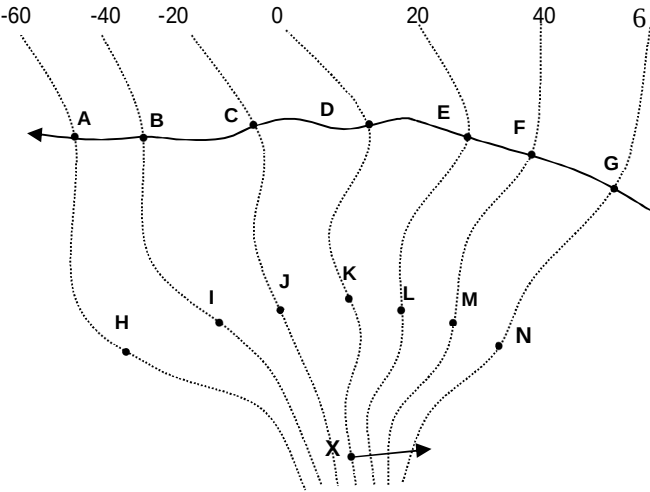


Question	Answer	Marks
4(a)	A region of space where a charged particle will experience an electric force.	1
(b)(i)	Look out for <u>both</u>	

	1. field line approximately perpendicular to equipotential lines at the points A, B, C, D, E, F, G. 2. direction of field line. From right to left (higher to lower potential) 	1 1
(b)(ii)	Relative field strength (within the same diagram) is represented by the density of equipotential lines. More/closely spaced equipotential lines represent a stronger field as compared to regions where the equipotential lines are more sparse. Hence the electric field is stronger at X.	1
(b)(iii)	Electric potential energy of $-5.0\mu\text{C}$ at L, $U = Vq = (20)(-5.0 \times 10^{-6}) = -1.0 \times 10^{-4} \text{ J}$	1 1
(b)(iv)	work required by external force (= - work done by the field) = $(V_f - V_i)q$ $\Rightarrow -1.0 \times 10^{-4} = (V_f - 20)(-5.0 \times 10^{-6})$ $\Rightarrow V_f = 40 \text{ V}$ Thus, possible points are F & M.	1 1
(b)(v)	$F = qE = q \frac{\Delta V}{\Delta x} = (-15.0 \times 10^{-6}) \left(\frac{20 - (-20)}{0.01} \right) = -0.60 \text{ N (1s.f)}$ (-ve sign indicates F opposite direction to E, which was from right to left. Thus general direction of F towards the right, drawn perpendicular to potential line) (Note that the caption in Fig 4.1 states “drawn to scale”. Thus accurate measurement of the distance is required)	1 1 1
	Max marks :	11

Question	Answer	Marks
5.(c)(vi)	Reading off graph, at 22.5°C	1